



Keynesian Analysis: Aggregate Expenditure

SESSION 4

<https://www.weforum.org/stories/2020/03/india-stimulus-support-lockdown-pandemic-covid19-epidemic-economics/>

India outlines \$23 billion stimulus to help poor hit by lockdown

Mar 27, 2020

India announced on Thursday a \$22.6 billion economic stimulus plan that provides direct cash transfers and food security measures, offering relief to millions of poor people hit by a nationwide lockdown triggered by the coronavirus pandemic.

The announcement confirms a Reuters report that the government had agreed on an economic stimulus package of more than \$20 billion plus to fight the coronavirus downturn.

<https://www.imf.org/external/pubs/ft/fandd/2010/12/blinder.htm>

Stimulus Worked

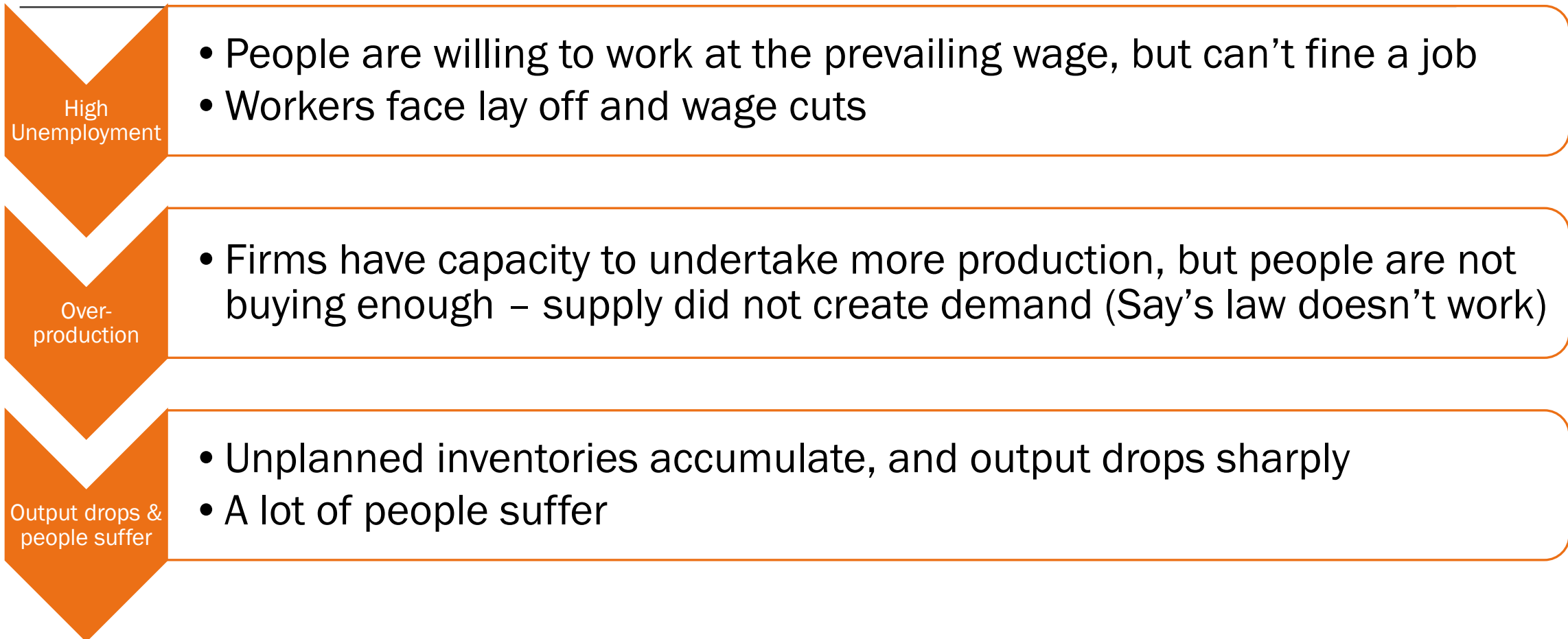
FINANCE & DEVELOPMENT, December 2010, Vol. 47, No. 4

Alan S. Blinder and Mark Zandi

THE U.S. economy has come a long way since the dark days of the Great Recession. Less than two years ago, the global financial system was on the brink of collapse, and the United States was suffering its worst economic downturn since the 1930s. At its worst, real gross domestic product (GDP) appeared to be in free fall, declining at nearly a 7 percent annual rate, with job losses averaging close to 750,000 a month. Today, the financial system is operating much more normally, real GDP has grown by more than 3 percent during the past year, and job growth has resumed, although at an insufficient pace.

From the perspective, say, of early 2009, this rapid turnabout was a surprise. Maybe the country and the world were just lucky. But we take another view: the Great Recession in the United States gave way to recovery as quickly as it did largely because of the unprecedented responses by monetary and fiscal policymakers.

Depressions and recessions are puzzling



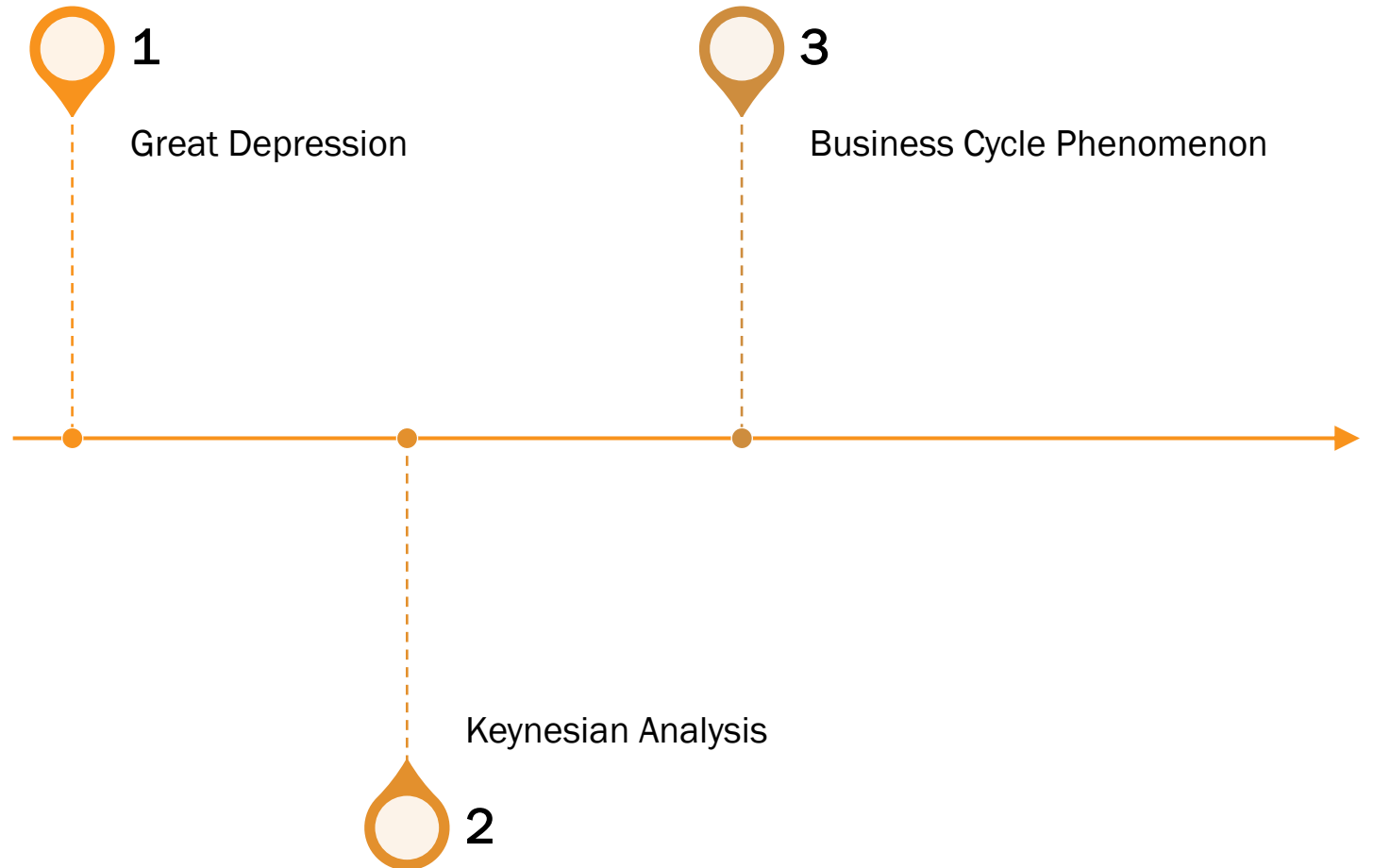
Historical evolution of ideas in Macroeconomics

Adam Smith – An Enquiry into the Nature and Causes of Wealth of Nations (1776)

J.M. Keynes – General Theory of Employment and Interest and Money (1936)

AS-AD Model

Outline



Event of Great Depression – 1930s



Stock market crash in US in 1929 – prices dipped by 90% (July 1932 - Sept 1929)

Millions of Americans lost their investment – fall in output, employment and prices

Took over a decade for the economy to recover

Brought a revolution in thinking on how economies work

TABLE 19-1 Economic Statistics of the Great Depression

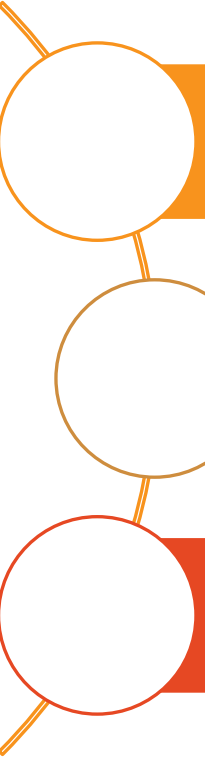
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	GNP,		G,	UN-		COM-				FULL
	1992 \$,	I/GNP,	1992 \$,	EMPLOY-	CPI,	MERCIAL	AAA	STOCK	MI	EMPLOY-
	BILLIONS	%	BILLIONS	MENT	1929	PAPER	RATE,	MARKET	1929	MENT
YEAR				RATE, %	= 100	RATE, %	%	INDEX*	= 100	SURPLUS/Y*, %†
1929	938.1	17.8	121.9	3.2	100.0	5.9	4.7	83.1	100.0	-0.8
1930	850.2	13.5	133.0	8.7	97.4	3.6	4.6	67.2	96.2	-1.4
1931	784.9	9.0	137.7	15.9	88.7	2.6	4.6	43.6	89.4	-3.1
1932	676.1	3.5	131.2	23.6	79.7	2.7	5.0	22.1	78.0	-0.9
1933	662.1	3.8	127.6	24.9	75.4	1.7	4.5	28.6	73.5	1.6
1934	713.7	5.5	145.2	21.7	78.0	1.0	4.0	31.4	81.4	0.2
1935	777.4	9.2	148.5	20.1	80.1	0.8	3.6	33.9	96.6	-0.1
1936	882.7	10.9	174.4	16.9	80.9	0.8	3.2	49.4	110.6	-1.1
1937	923.5	12.8	167.8	14.3	83.3	0.9	3.3	49.2	114.8	1.8
1938	885.7	8.1	182.7	19.0	82.3	0.8	3.2	36.7	115.9	0.6
1939	953.0	10.5	190.2	17.2	81.0	0.6	3.0	38.5	127.3	-0.1

*Stock market index is Standard & Poor's composite index, which includes 500 stocks; September 1929 = 100.

†Y* denotes full-employment output.

Source: Cols. 1, 2, 3: U.S. Department of Commerce, *The National Income and Product Accounts of the United States, 1929-1974*. Col. 4: Revised Bureau of Labor Statistics data taken from Michael Darby, "Three-and-a-Half Million U.S. Employees Have Been Mislaid: Or, an Explanation of Unemployment, 1934-1941," *Journal of Political Economy*, February 1976. Cols. 5, 6, 7: *Economic Report of the President, 1957*. Col. 8: Standard & Poor's Statistical Service, *Security Price Index Record, 1978*. Col. 9: Milton Friedman and Anna J. Schwartz, *A Monetary History of the United States, 1867-1960* (Princeton, NJ: Princeton University Press, 1963), table A1, col. 7. Col. 10: E. Cary Brown, "Fiscal Policy in the Thirties: A Reappraisal," *American Economic Review*, December 1956, table 1, cols. 3, 5, and 19.

Prior to Great Depression



Flexible prices adjust supply and demand mismatch

Say's Law of Market – full employment of resources and no scope for over production

Dominance of supply side factors

Keynesian Revolution

- Explained forces working behind Great Depression like situation
- Dominance of effective demand (aggregate spending) and sticky wages (prices)
- Great depression was caused due to deficient demand
- Role of public policy to improve employment situation and economic activities

Great Moderation (since 1980s)

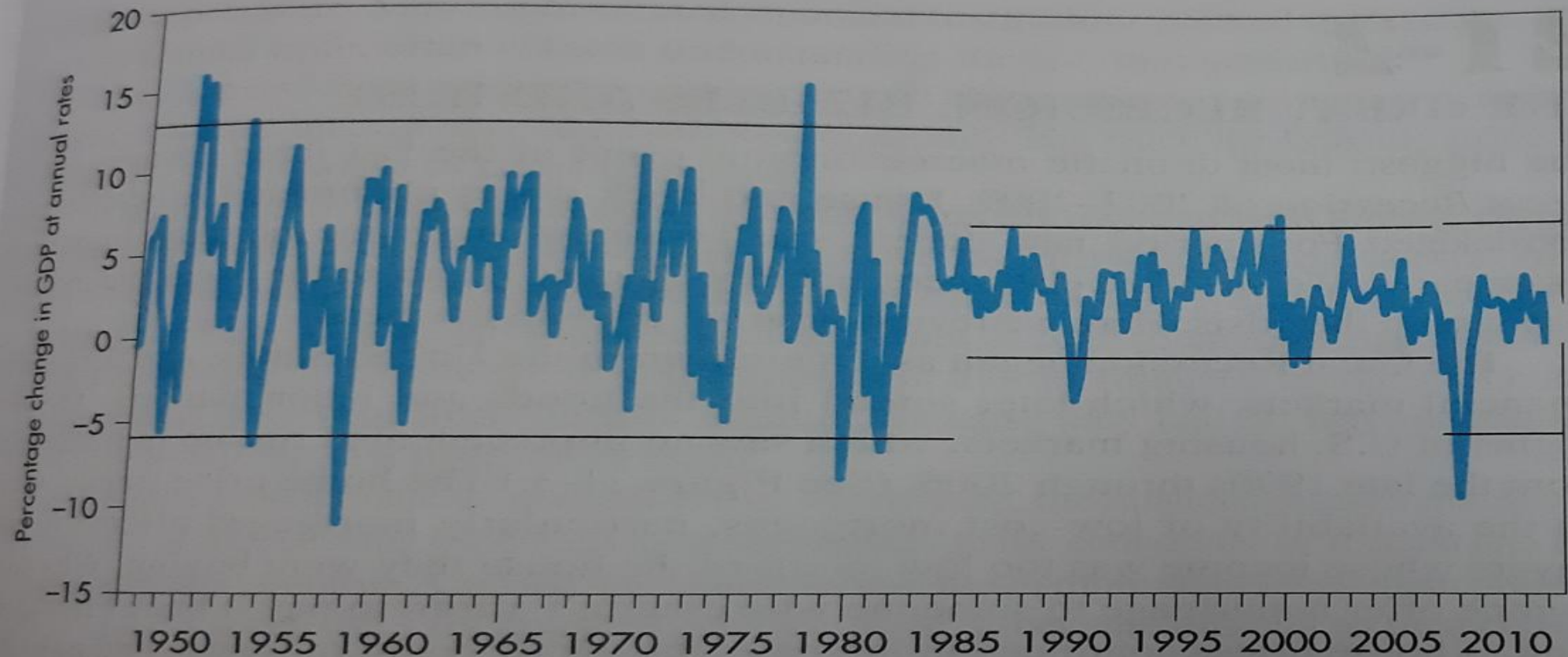
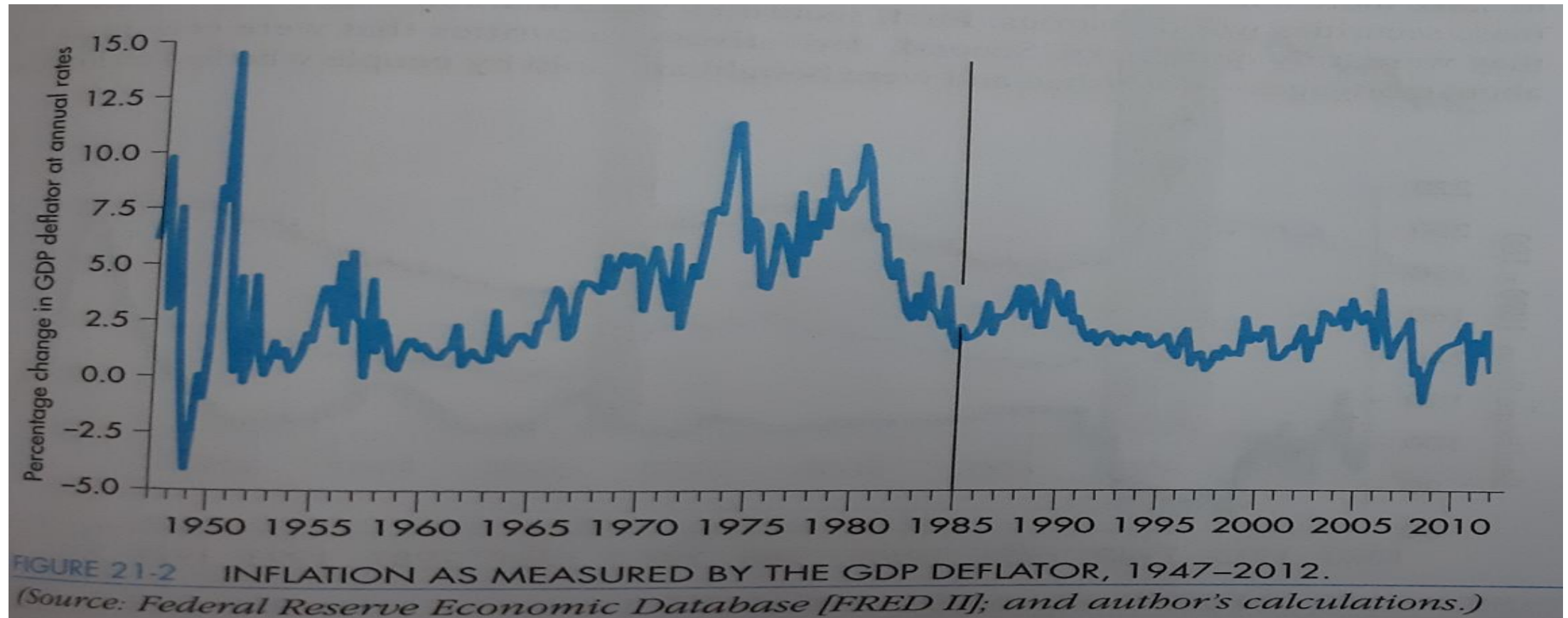
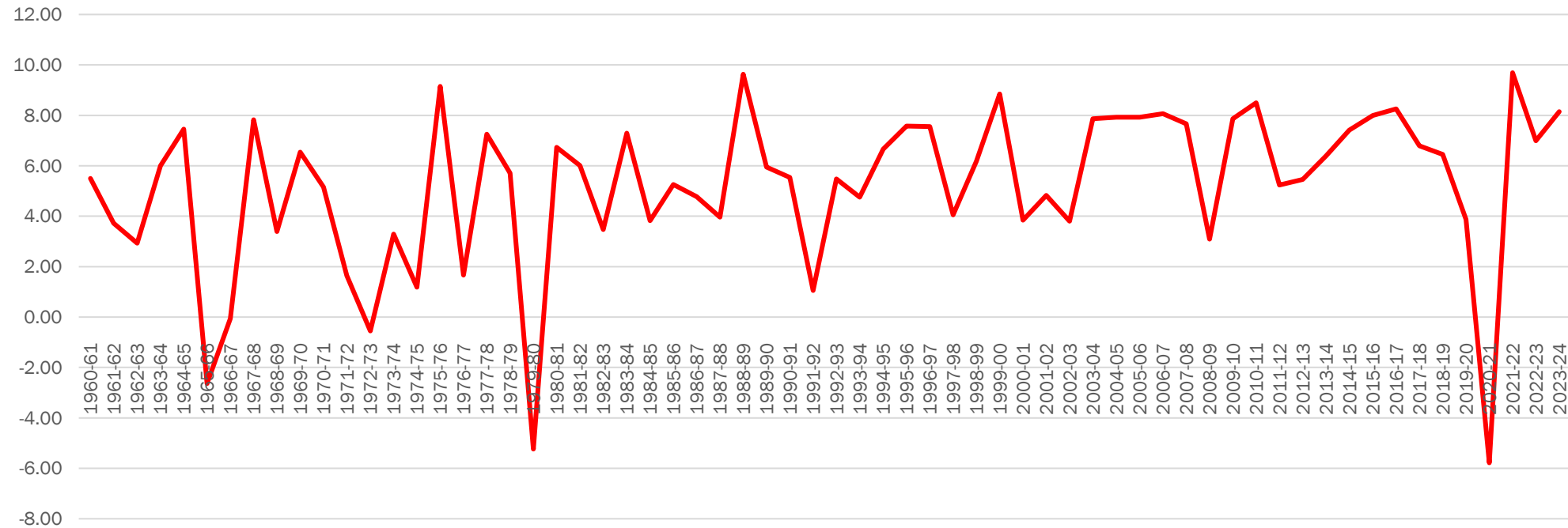


FIGURE 21-1 QUARTERLY GDP GROWTH AT ANNUAL RATES, 1947–2012.

(Source: Federal Reserve Economic Database [FRED II]; and author's calculations.)



Indian perspective – GDP Growth Rate



Keynesian Analysis

Components of Aggregate Demand

$$Y = C + I + G + NX$$

1. Consumption (C)
2. Investment (I)
3. Government Expenditure (G)
4. Net Exports (NX)

TABLE 4 : COMPONENTS OF GROSS DOMESTIC PRODUCT**(Base Year : 2011-12) Constant Prices**

							(Amount in ₹ Crore)
Item/Year	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22
1	2	3	4	5	6	7	8
Private Final Consumption Expenditure	6381419	6900236	7330728	7850444	8259704	7763734	8377854
Government Final Consumption Expenditure	1132802	1201598	1344843	1434945	1484272	1537603	1577132
Gross Fixed Capital Formation	3492183	3787568	4083079	4540509	4611021	4131279	4784054
Changes in Stocks	239557	122639	206436	262771	108284	-11573	187937
Exports of Goods and Services	2370282	2488423	2602012	2912480	2813609	2553683	3174844
Import of Goods and Services	2511540	2621593	3078274	3349861	3321586	2862871	3878193
Discrepancies	78804	277844	443451	149920	396125	238638	216842
Gross Domestic Product	11369493	12308193	13144582	13992914	14515958	13558473	14735515

Share of Various Components of Aggregate Expenditure - India

Item	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22
Consumption Expenditure	56.1	56.1	55.8	56.1	56.9	57.3	56.9
Government Expenditure	10.0	9.8	10.2	10.3	10.2	11.3	10.7
Investment	32.8	31.8	32.6	34.3	32.5	30.4	33.7
Net Exports	-1.2	-1.1	-3.6	-3.1	-3.5	-2.3	-4.8
GDP	100.0	100.0	100.0	100.0	100.0	100.0	100.0

Consumption Expenditure

Most dominant component of aggregate spending

We make consumption choices everyday - Aware about the factors determine our choices

1. Current income – MPC (0.6) – (Permanent)
2. Wealth
3. Expected future income
4. Interest rate on savings and borrowing

MPC

It refers to the amount of increase in current consumption expenditure due to one unit increase by disposable income

Example:

- When income was Rs.10,000, say consumption was Rs.6,000, and when income increased to Rs.11,000, consumption increased to Rs.6,600
- Thus, increase in disposable income by Rs.1,000 led to increase in consumption expenditure by Rs.600
- $MPC = \Delta C / \Delta DY = 600 / 1000 = 0.6$

Investment

Investment for layman (buying FD or equity) *versus* economist (change in capital – machines, factories, buildings, software, residential houses, ***change in inventories***)

Determinants:

1. Expected Profitability
2. Interest rate (compare with MEC or IRR)
3. Business (corporate) taxes

Government expenditure

Driven by norms for provision of certain public goods like education, health, roads and bridges, national security, law and order, etc.

(Also to moderate economic fluctuations through appropriate fiscal policy – to stimulate or restrain the economy)

Not affected directly by standard macroeconomic factors such as aggregate income, wealth, interest rate, etc.

Net exports

Exports over imports

Determinants:

1. Domestic income
2. Foreign income
3. Foreign exchange rate
4. Tastes for foreign goods
5. Policies related to international trade

TABLE 11-4**Summarizing all determinants of aggregate expenditure**

This chart summarizes Tables 11-1 through 11-3. It shows at a glance how increases in the determinants of aggregate expenditure directly affect all components of aggregate expenditure. We can usually predict an increase or decrease.

Increase in determinant	Effect on C	Effect on I	Effect on G	Effect on NX	Effect on aggregate expenditure (Y)
Domestic income	Increase	—	—	Decrease	Generally increase
Wealth	Increase	—	—	—	Increase
Expected future income	Increase	—	—	—	Increase
Interest rate	Decrease	Decrease	—	—	Decrease
Expected profitability	—	Increase	—	—	Increase
Business taxes	—	Decrease	—	—	Decrease
Foreign income	—	—	—	Increase	Increase
Exchange rates	—	—	—	Decrease	Decrease
Tastes for foreign goods	—	—	—	Decrease	Decrease
Trade policies	—	—	—	Depends	Depends

Keynesian Equilibrium and income determination

Relevance



Prior to Great Depression of 1930s, it was believed that the only equilibrium for the economy is at full-employment level

In the backdrop of the Great Depression, Keynesian General Theory suggested that equilibrium can occur, when planned expenditure is equal to actual expenditure

It is not necessary that the equilibrium has to occur only at the full-employment level – multiple equilibria

When the equilibrium is different from full-employment level, there is a need for public policy intervention to bring it closer to the full-employment level

Actual *versus* Planned Expenditure

Actual Expenditure

- Expenditure actually incurred in the form of **C**, **I**, **G** and **NX**
- $GDP = C + I + G + NX$

Planned Expenditure

- Amount of spending and production that businesses, households, and others planning to make
- Sum of planned **C**, **I**, **G** and **NX**

Planned Investment



Planned investment by the firms depends on the expectations on sales in the future, interest rates and corporate taxes

After assessing future business prospects and taking into account existing interest rates and taxes, firms decide how much to invest

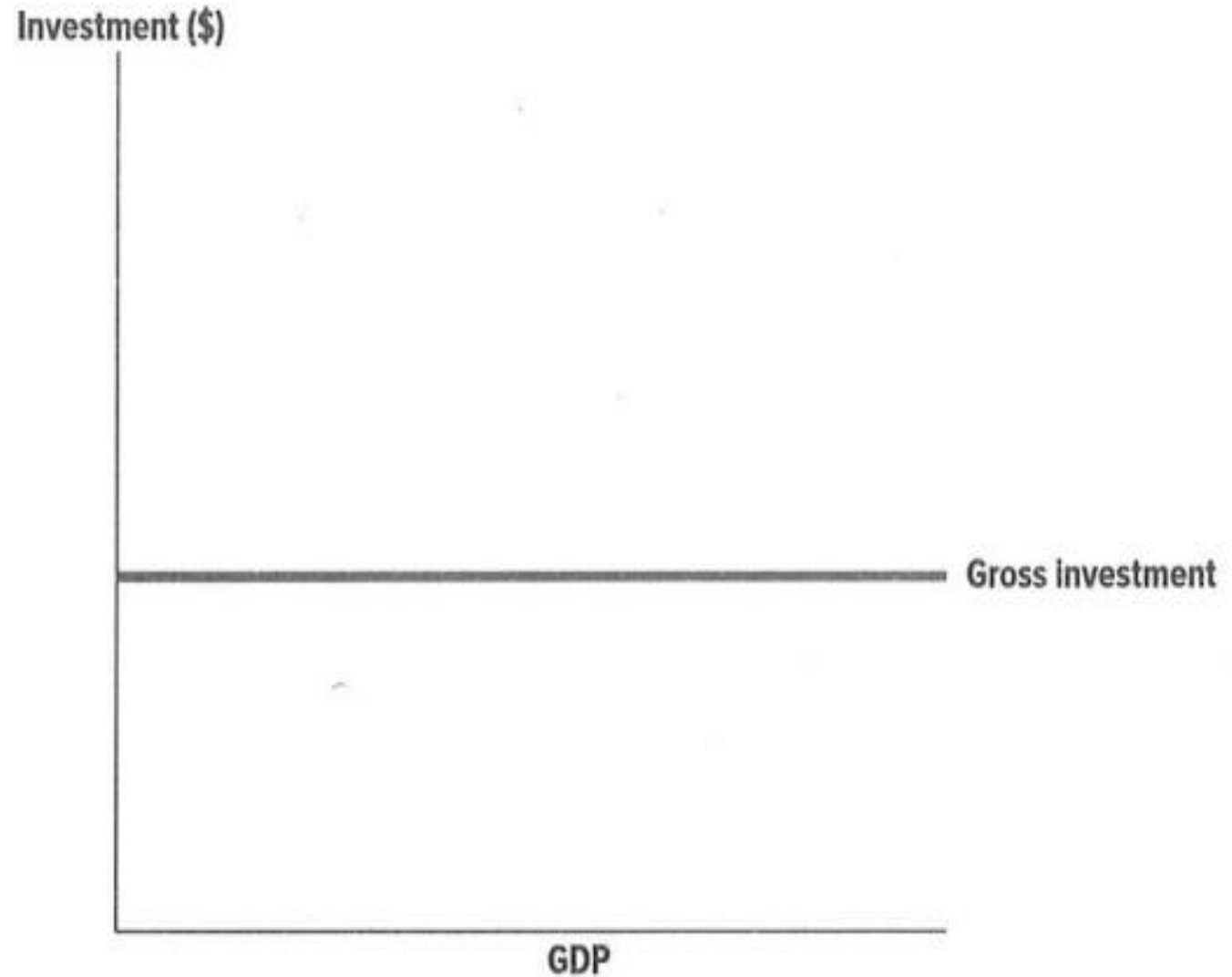
This planned investment is independent of actual GDP (I^p)

Planned inventories does not match with actual inventories, if aggregate spending differs from aggregate output

FIGURE 11-2

The investment

schedule Investment spending is the same for each level of GDP.



Planned consumption spending

Autonomous consumption spending

- Does not depend on income (**a**)
- Rent for housing, EMI for car loan, etc.
- Subsistence food consumption

Induced consumption spending

- Depends on current (**disposable**) income (**bY**)
- Spending on eating out, buy luxury items, holiday trips, etc.

Planned consumption spending

- **$C = a + bY$**
- **b**: *marginal propensity to consume*; $0 < b < 1$

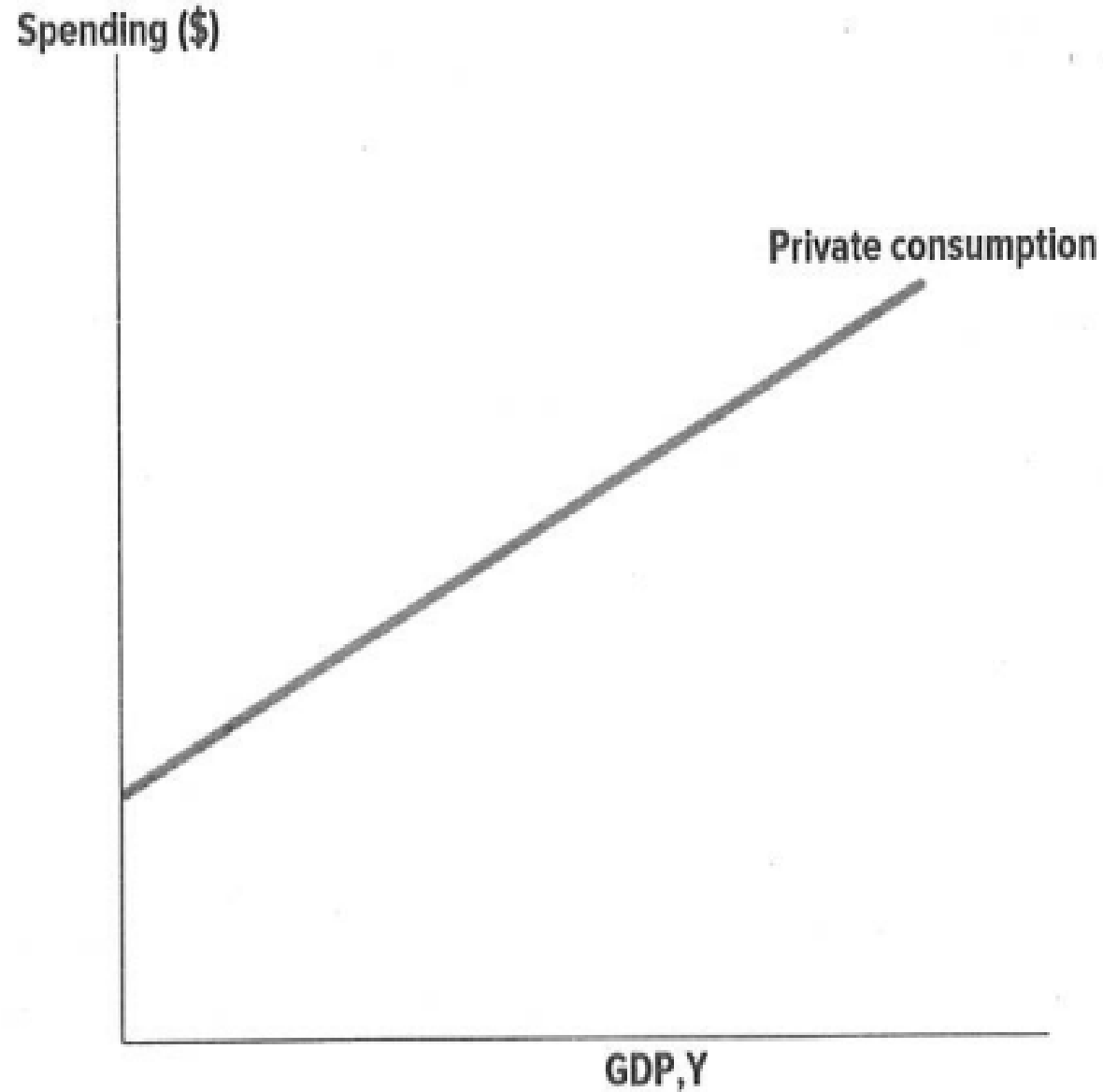
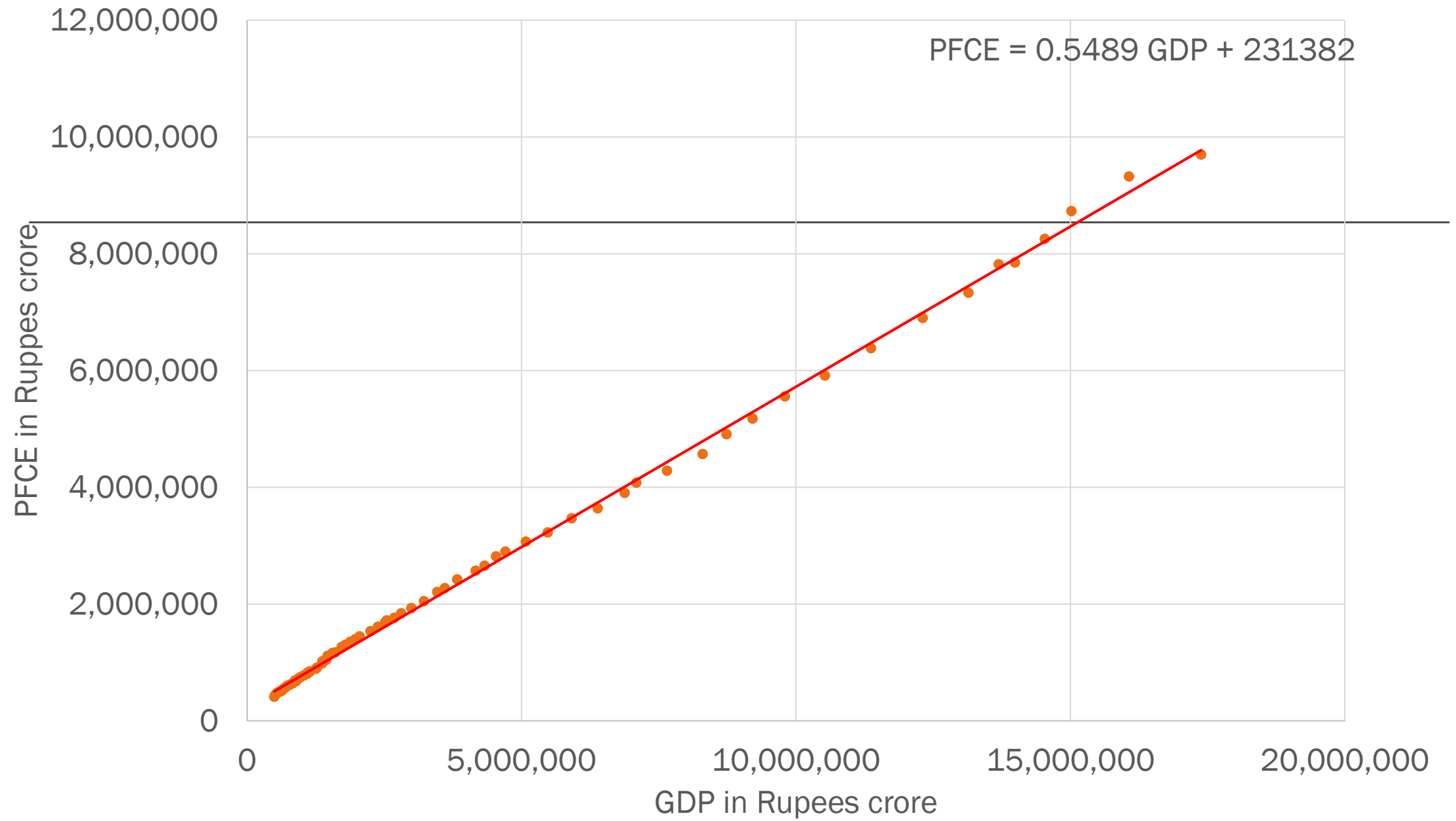


FIGURE 11-3
Consumption and GDP Consumption spending is made up of two parts: autonomous spending, which occurs independent of GDP, and spending that increases as GDP increases.



Planned government spending and Net exports

Government
spending
(Autonomous)

- Planned government spending does not depend on current GDP (G^e)

Net exports
(Autonomous)

- Planned exports and imports depend on expected foreign/domestic economic conditions, tastes and preferences, exch. rate, trade policy
- Assumed to be autonomous for simple analysis (NX^e)

Equality between Planned Agg Expenditure & Actual

$$PAE = (a+bY) + I^e + G^e + NX^e$$

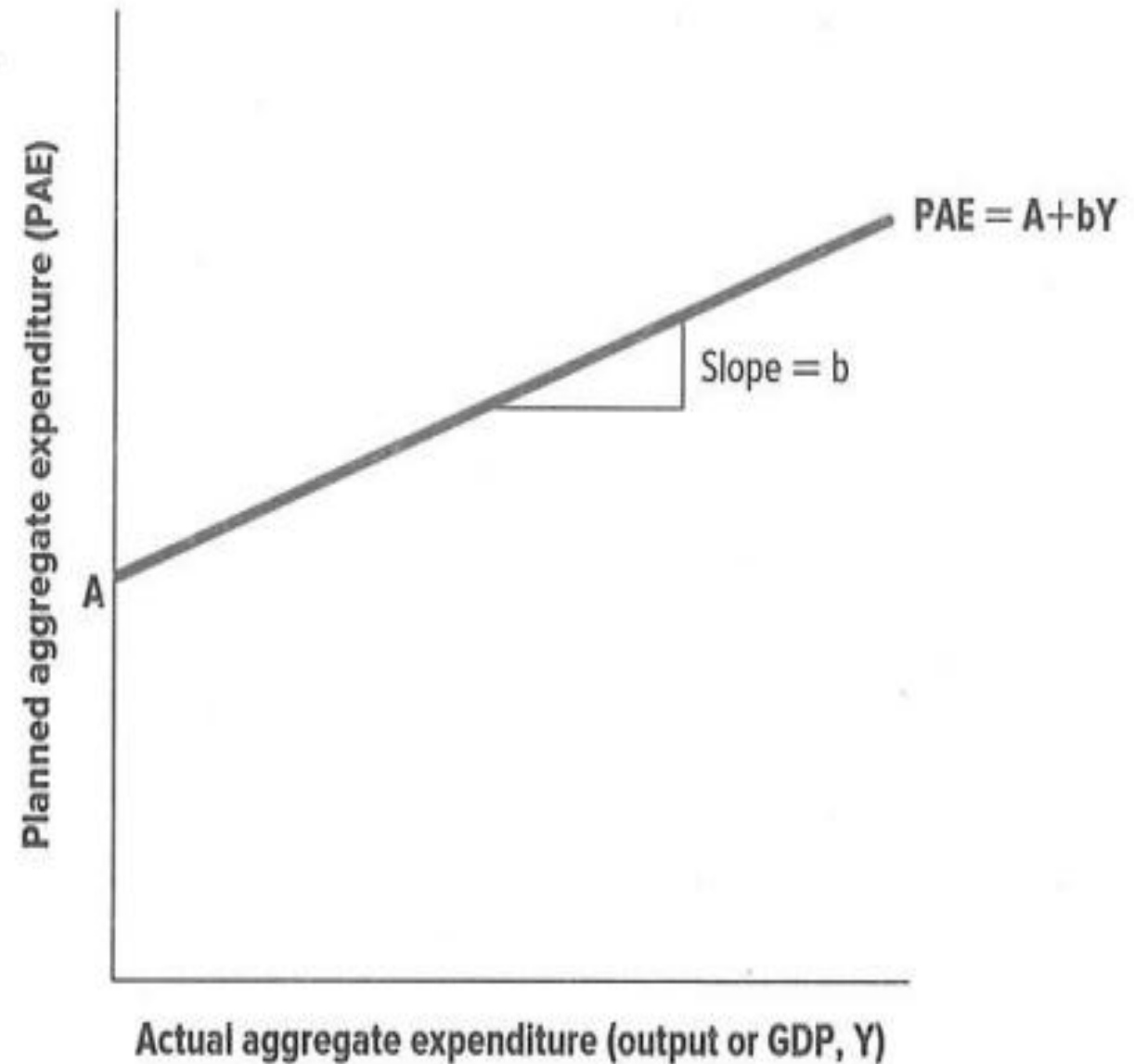
$$PAE = A + bY, \text{ where } A = a + I^e + G^e + NX^e$$

Equilibrium condition: $PAE=Y$

FIGURE 11-4

Planned aggregate expenditure

This figure shows planned aggregate expenditure as a function of income.



Keynesian equilibrium

Insight

- Firms don't produce at the maximum capacity
- They produce what they can sell

Sub-capacity

- Under weak demand, firms will produce less than capacity

Equilibrium

- $\text{Planned Agg Exp} = \text{Actual Agg Exp} = Y$
- Any deviation leads to unplanned changes in inventories & production

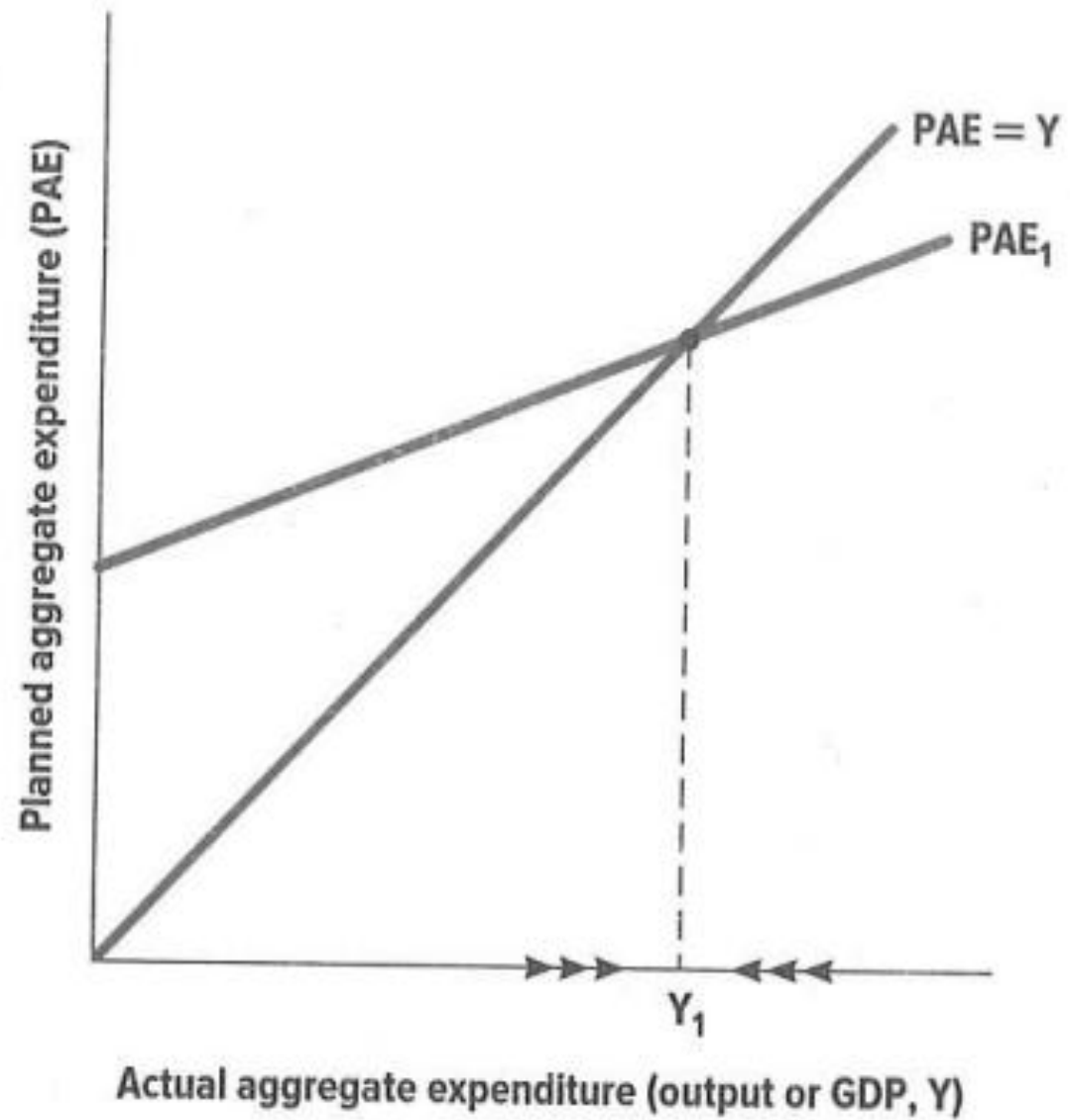
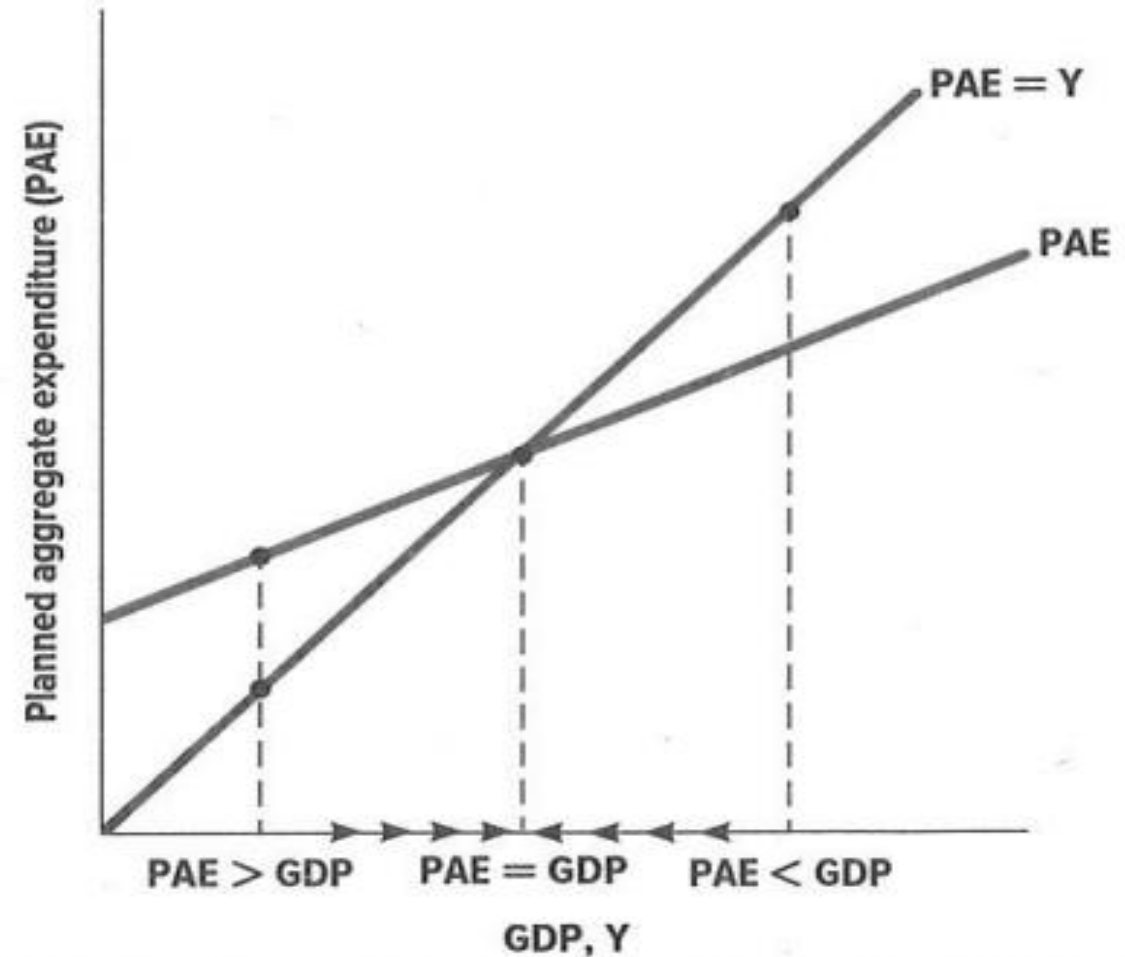


FIGURE 11-5
The Keynesian cross

FIGURE 11-6

Disequilibrium in the Keynesian cross



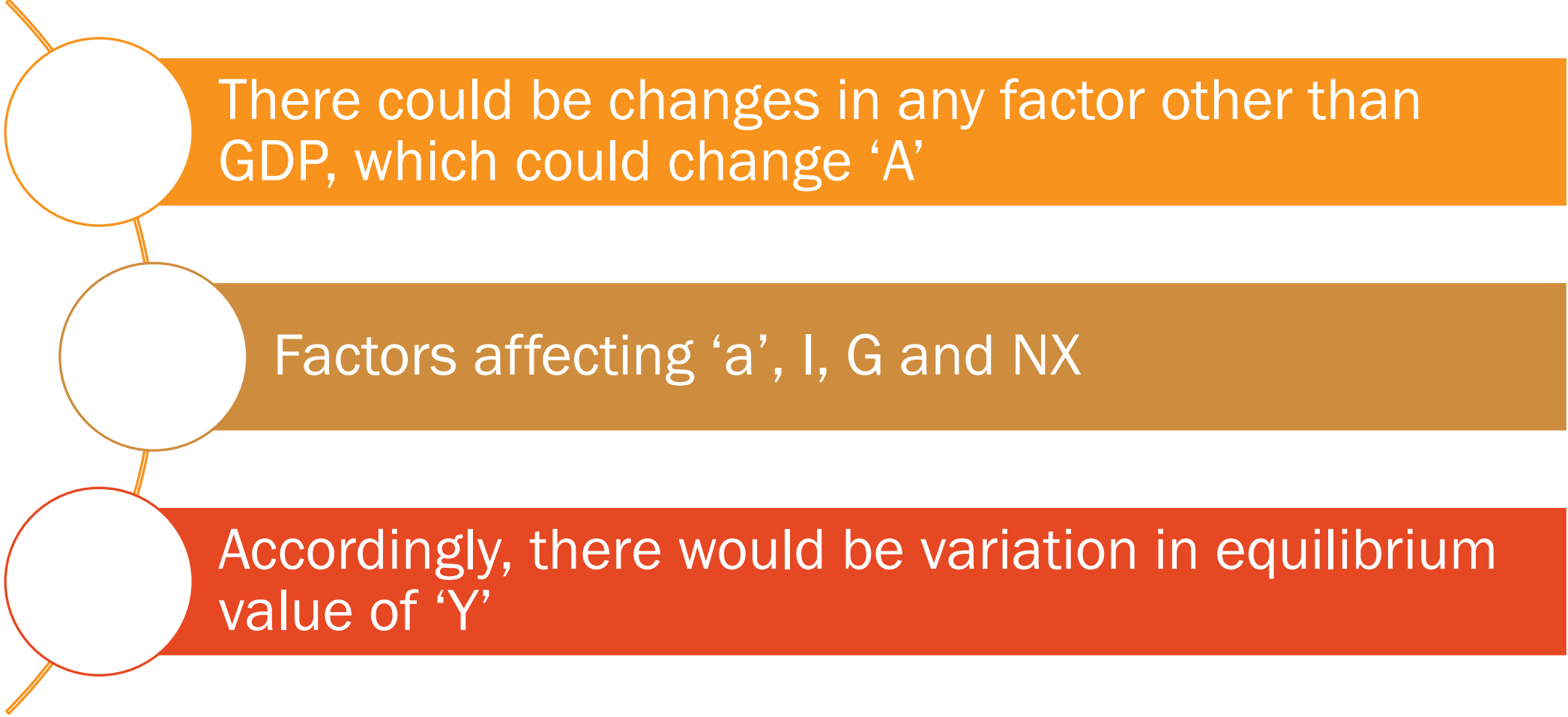
The left side of the graph shows planned aggregate expenditure higher than GDP. At this point, firms lose inventory and increase production. On the right side of the graph, GDP is above planned aggregate expenditure, leading firms to cut production.

TABLE 11-5

Finding equilibrium aggregate expenditure (in trillions of dollars)

Real GDP	Consumption	Investment	Government spending	Net exports	Aggregate expenditure	Unplanned change in inventories	Production decision
17	12.5	4	3	-0.5	19	-2	Increase
18	13	4	3	-0.5	19.5	-1.5	Increase
19	13.5	4	3	-0.5	20	-1	Increase
20	14	4	3	-0.5	20.5	-0.5	Increase
21	14.5	4	3	-0.5	21	0	Equilibrium
22	15	4	3	-0.5	21.5	0.5	Decrease
23	15.5	4	3	-0.5	22	1	Decrease
24	16	4	3	-0.5	22.5	1.5	Decrease

Shifting of PAE



There could be changes in any factor other than GDP, which could change 'A'

Factors affecting 'a', I, G and NX

Accordingly, there would be variation in equilibrium value of 'Y'

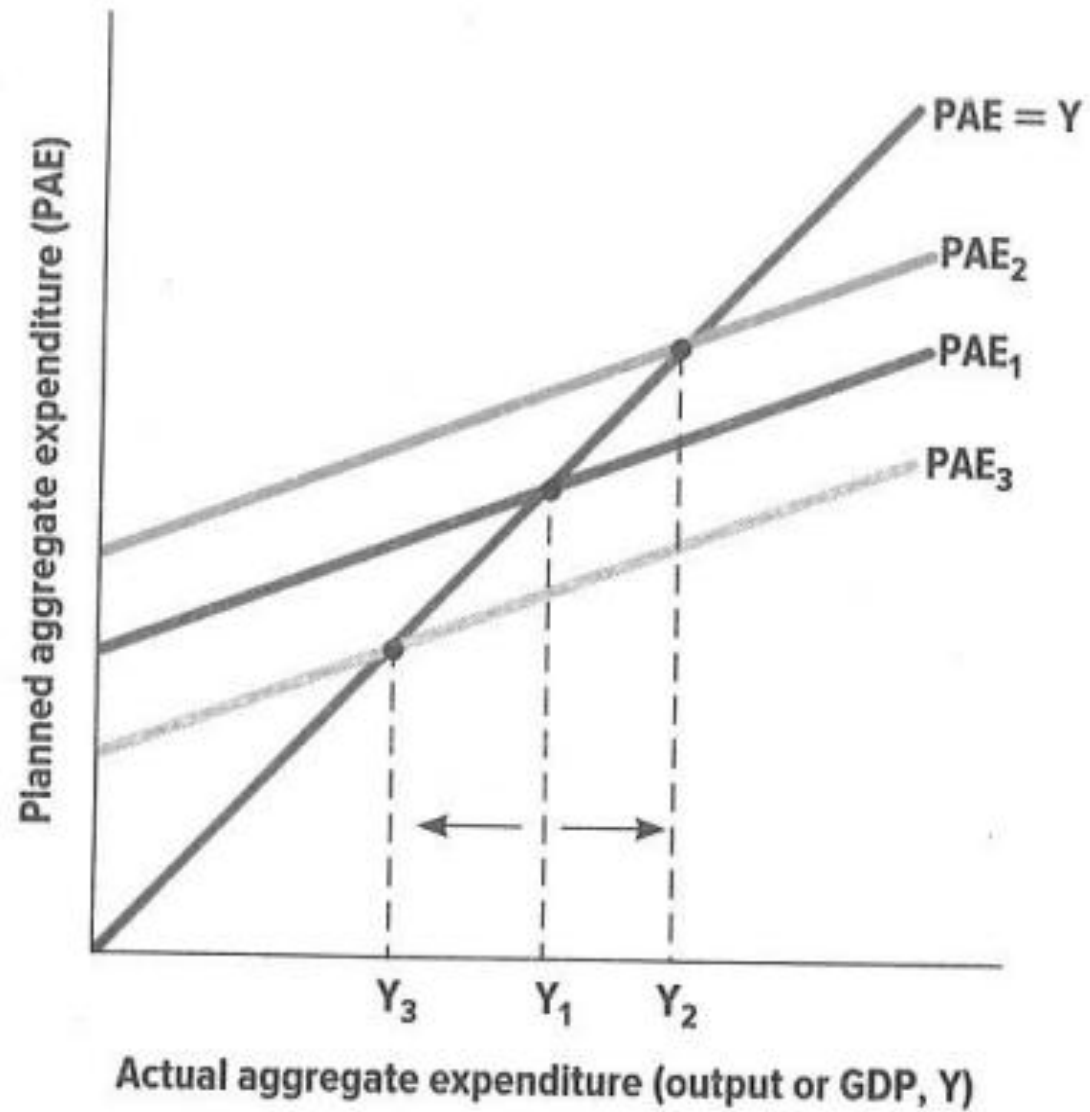


FIGURE 11-7
Nonincome
determinants and
changes in aggregate
expenditure
equilibrium

Numerical example 1:

A Simple Keynesian Model:

- Consumption expenditure: $C = 30 + 0.75 (Y - T)$
- $I_{\text{planned}} = 50$
- $G = 100$
- $NX = 30$
- $T = 100$

Based on the above, calculate equilibrium output (GDP)

Numerical example 1: Solution

A Simple Keynesian Model:

At equilibrium, $PAE = Y$

$$PAE = Y_{\text{planned}} = 30 + 0.75(Y - 100) + 50 + 100 + 30 = 135 + 0.75Y$$

$$PAE = Y \text{ implies that } Y = 135 + 0.75Y$$

$$Y = (1/0.25) * 135 = 4 * 135$$

Thus, $Y^* = 540$

Numerical example 2:

A Simple Keynesian Model:

- Consumption expenditure: $C = 30 + 0.75 (Y - T)$
- $I_{\text{planned}} = 50$
- $G = 200$
- $NX = 30$
- $T = 200$

Based on the above, calculate equilibrium output (GDP)

Numerical example 2: Solution

A Simple Keynesian Model:

At equilibrium, $PAE = Y$

$$PAE = Y_{\text{planned}} = 30 + 0.75 (Y - 200) + 50 + 200 + 30 = 160 + 0.75 Y$$

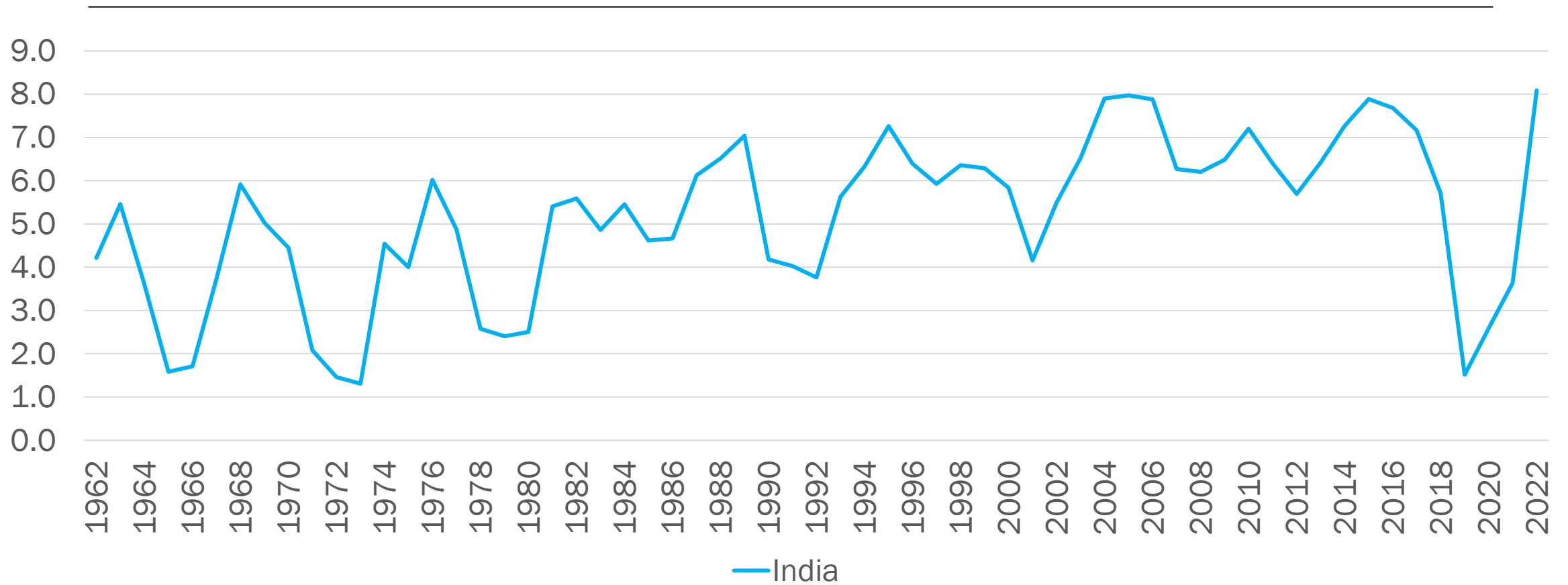
$$PAE = Y \text{ implies that } Y = 160 + 0.75 Y$$

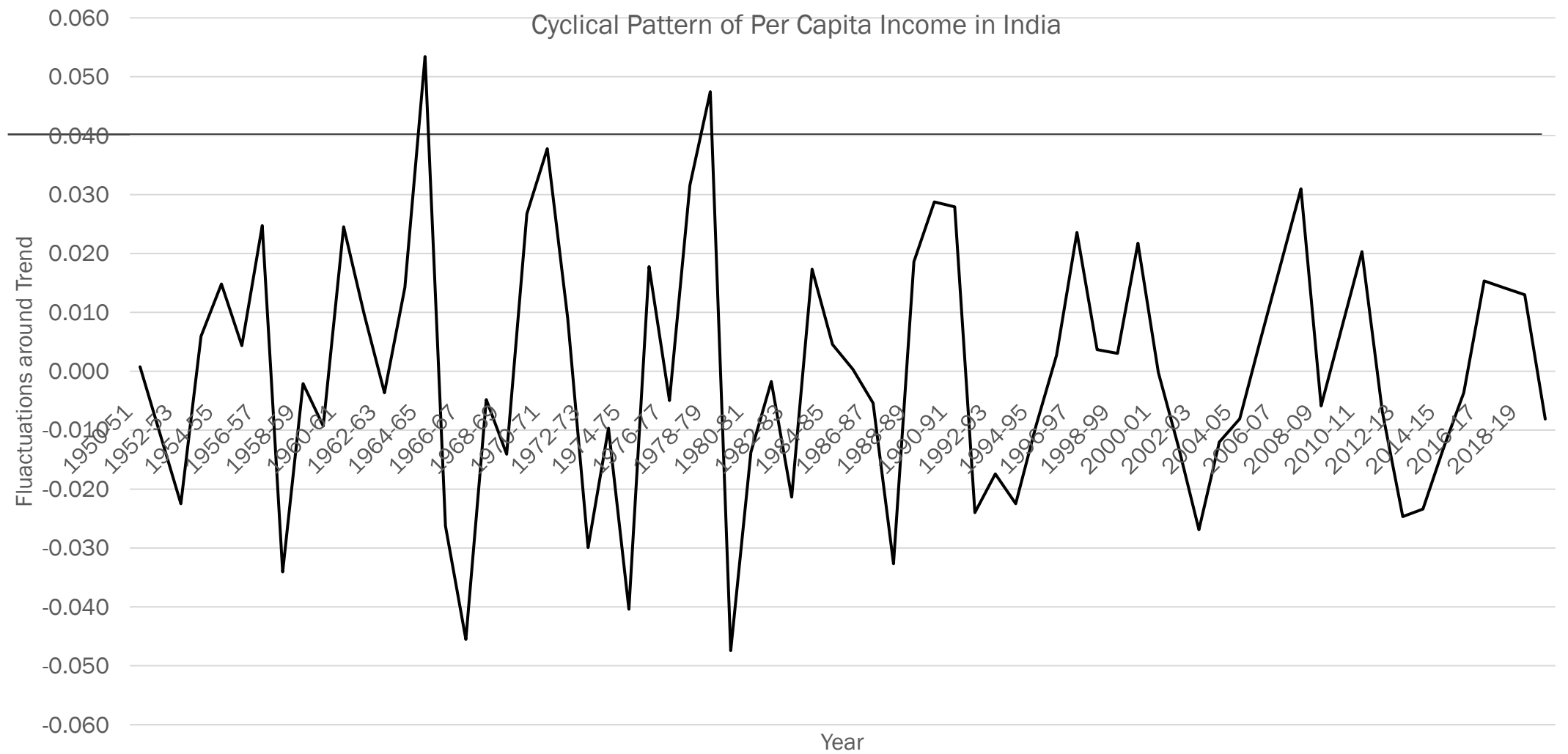
$$Y = (1/0.25) * 160 = 4 * 160$$

Thus, $Y^* = 640$

Business Cycle Phenomenon

Economic Growth (3 Year MA)





Business cycle

Boom

- Actual output (GDP) is higher than potential output
- Low unemployment rate
- Generally associated with high inflation

Recession

- Actual output is lower than potential output
- High unemployment rate
- Generally associated with disinflation/deflation

Output Gap

Output gap = Actual Output – Potential Output

Potential output is the level produced with full-employment

No cyclical unemployment - NAIRU

FIGURE 11-8

A recessionary output gap

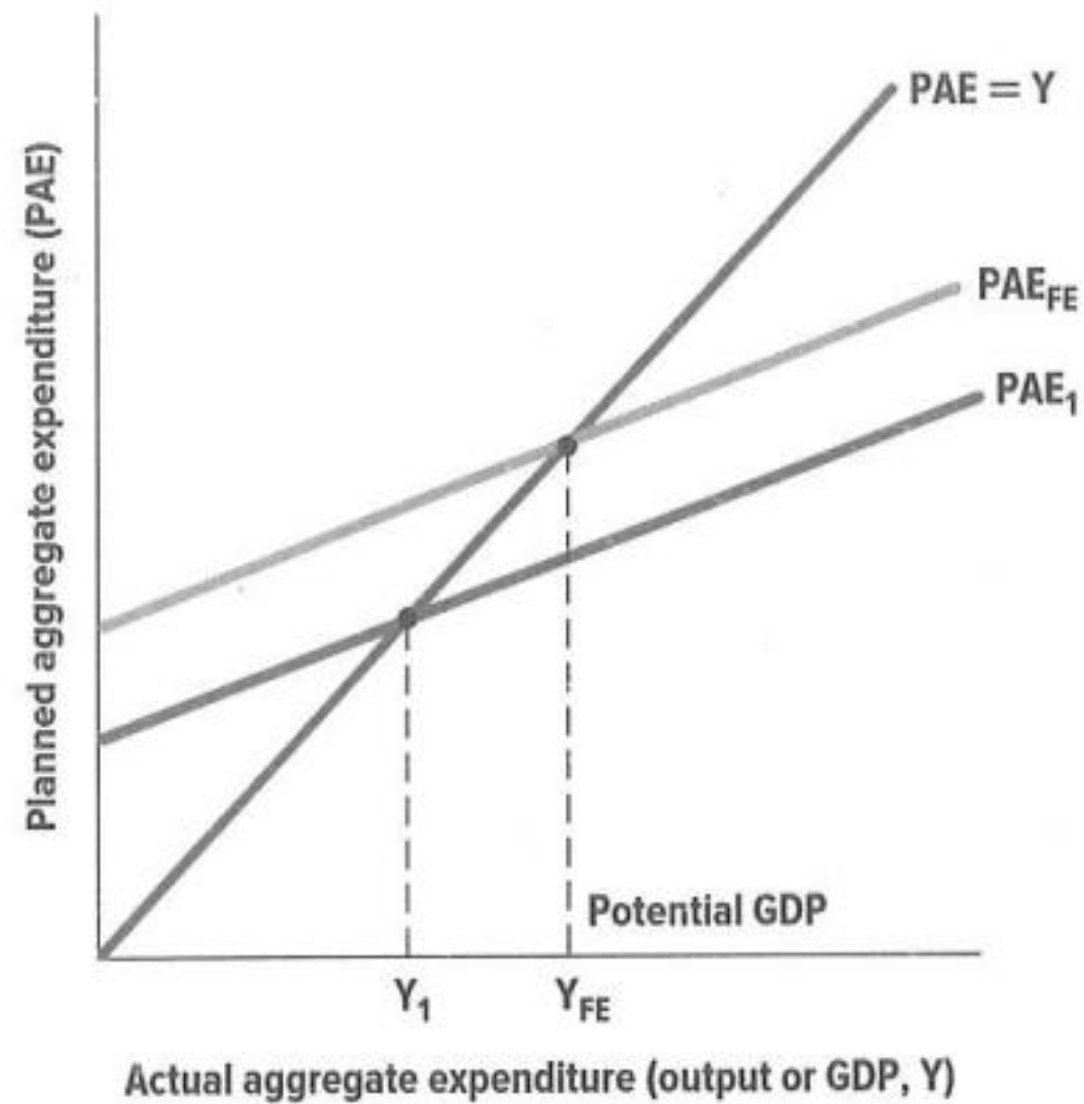
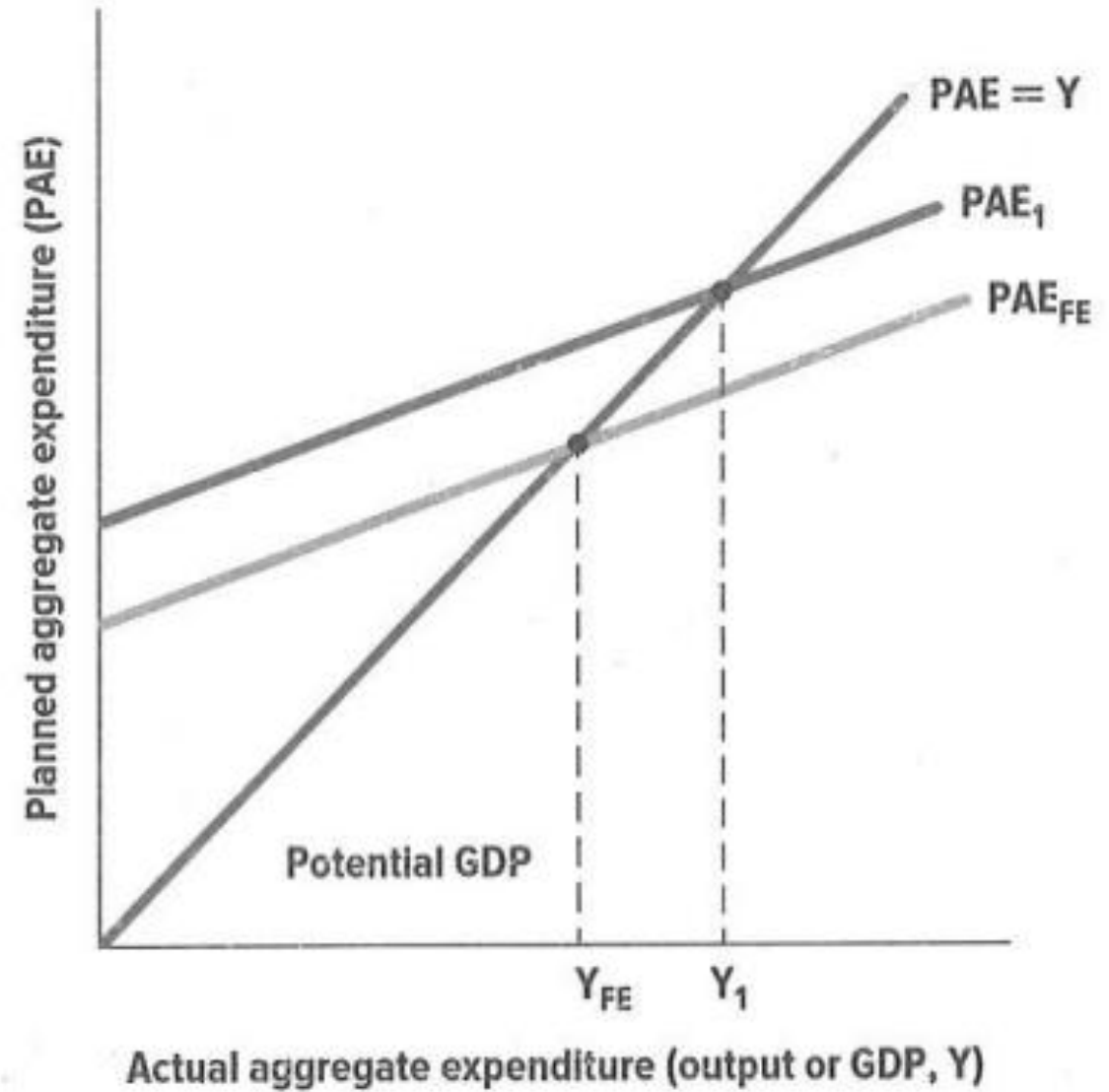


FIGURE 11-9
An inflationary output gap



Autonomous Spending Multiplier

From the previous analysis ...

It appears that rise in government expenditure (G) will raise aggregate demand (AD) – shifting the AD curve in Keynesian cross to a higher level and the economy will produce a higher level of Y – at a new equilibrium.

The question is: If G is raised by say, INR 1,000 crore, the new equilibrium Y will be how much higher as compared to the old equilibrium Y?

Answer is: $\Delta Y > \Delta G$

(ignoring funding of rise in G)

It can be explained by taking into account a series of effects

First – rise in G raises Y by the same amount (Why)

Second – rise in Y raises C, and then Y rises again

Successive impact if rise in Y leading to further rise in C and Y continues

TABLE 9-1 The Multiplier

ROUND	INCREASE IN DEMAND THIS ROUND	INCREASE IN PRODUCTION THIS ROUND	TOTAL INCREASE IN INCOME (ALL ROUNDS)
1	$\Delta \bar{A}$	$\Delta \bar{A}$	$\Delta \bar{A}$
2	$c\Delta \bar{A}$	$c\Delta \bar{A}$	$(1 + c)\Delta \bar{A}$
3	$c^2\Delta \bar{A}$	$c^2\Delta \bar{A}$	$(1 + c + c^2)\Delta \bar{A}$
4	$c^3\Delta \bar{A}$	$c^3\Delta \bar{A}$	$(1 + c + c^2 + c^3)\Delta \bar{A}$
...	...	...	...
...	...	...	...
...	...	...	$\frac{1}{1 - c}\Delta \bar{A}$

$$\begin{aligned}
 \Delta AD &= \Delta \bar{A} + c\Delta \bar{A} + c^2\Delta \bar{A} + c^3\Delta \bar{A} + \dots \\
 &= \Delta \bar{A}(1 + c + c^2 + c^3 + \dots)
 \end{aligned}
 \tag{15}$$

$$\Delta AD = \frac{1}{1 - c} \Delta \bar{A} = \Delta Y_0
 \tag{16}$$

$$\alpha \equiv \frac{1}{1 - c}
 \tag{17}$$

Multiplier

A given change in (initial) autonomous expenditure has ripple effects of several round

Final increase in aggregate expenditure will a multiple of initial increase

If $\Delta G=100$, in first round $\Delta Y=\Delta G$.

In second round, $\Delta C=b\Delta Y$. If $b=0.5$, then $\Delta Y=\Delta C=0.5*100 = 50$

So, after second round $\Delta Y=\Delta G + \Delta C = 100+50 = 150$, and so on.

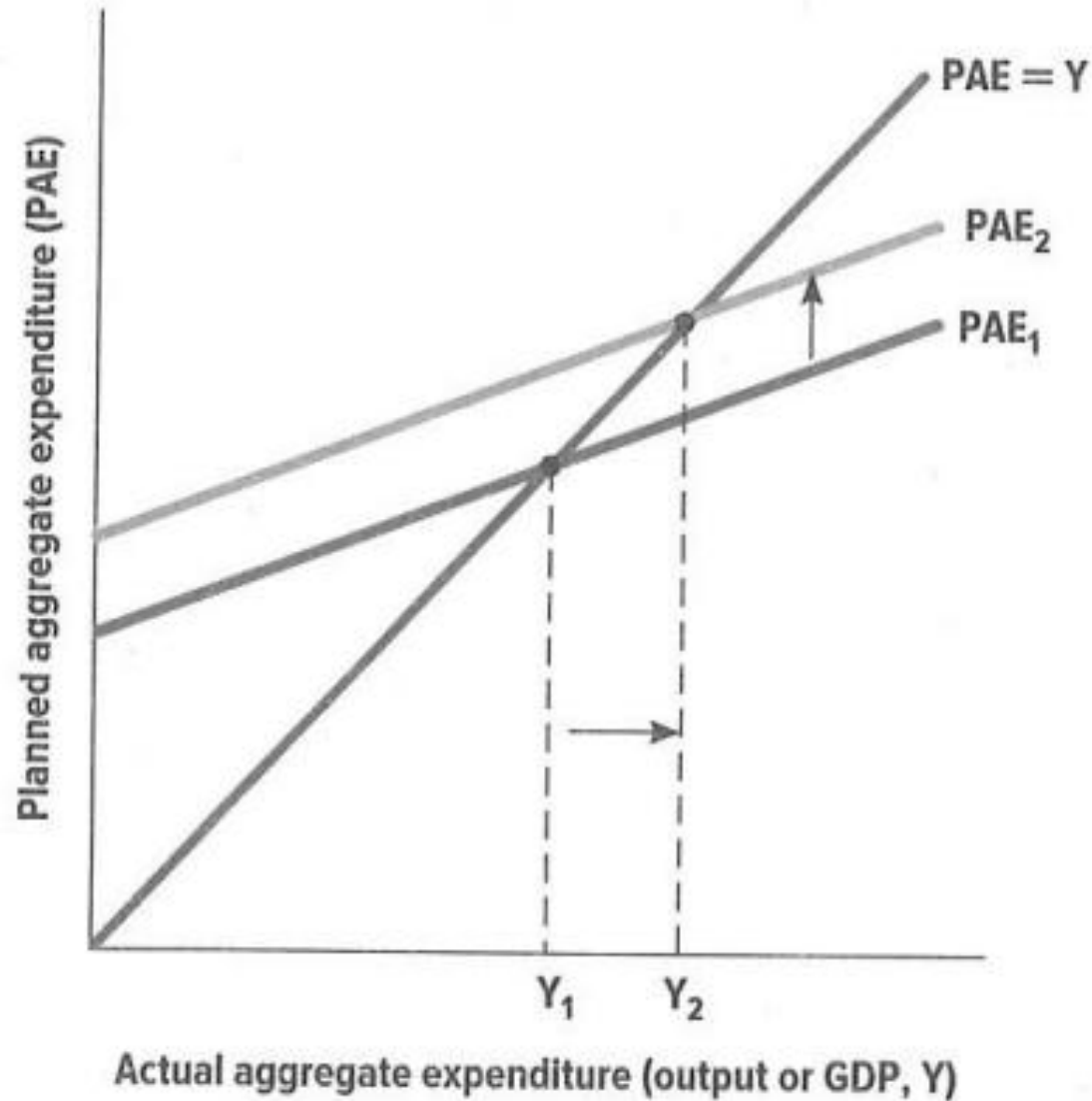
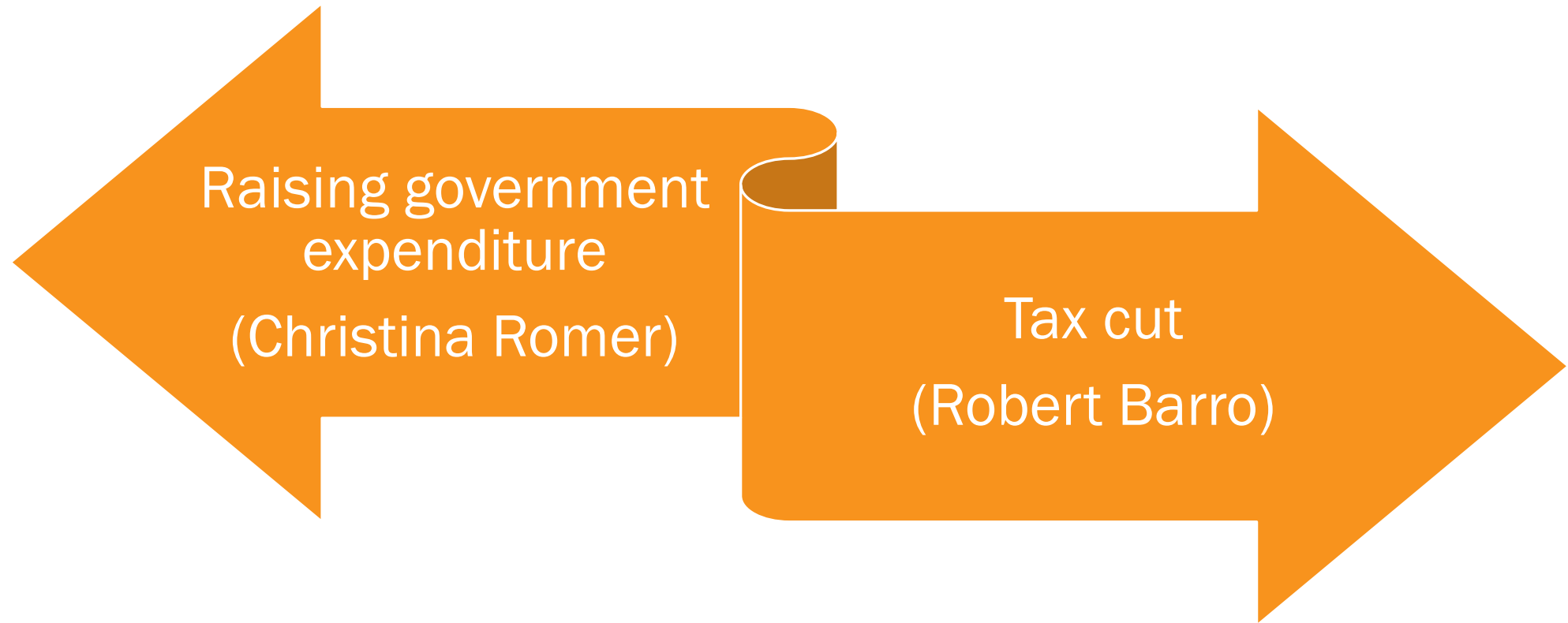


FIGURE 11-10
Overall change in
equilibrium aggregate
expenditure

The multiplier can be approximated by comparing the length of the horizontal arrow to the length of the vertical arrow. In the case drawn here, the multiplier is greater than 1, because the horizontal arrow is longer.

Policy dilemma (GFC)



Expenditure multiplier: Algebra 1

$$PAE = A + bY$$

(ignoring government tax)

At equilibrium, $PAE = Y$

$$\text{Thus, } Y = A + bY \quad \text{or} \quad Y - bY = A \quad \text{or} \quad Y = A/(1-b)$$

With change in A by ΔA , we get,

$$\Delta Y = [1/(1-b)] * (\Delta A) = m * (\Delta A)$$

Example: If $A=100$, and $b = 0.8$

$$\Delta Y = [1/(1-0.8)] * 100 = [1/0.2] * 100 = 5 * 100 = 500$$

$$\Delta Y = \$100 + b \times \$100 + b^2 \times \$100 + b^3 \times \$100 + \dots = \$100 \times (1 + b + b^2 + b^3 + \dots)$$

The sum in the parentheses is the sum of an infinite geometric series, which for $b < 1$ is simply equal to $1/(1 - b)$. Thus, the final change in output above is equal to

EQUATION 11-4

$$\Delta Y = \$100 \times \frac{1}{1 - b}$$

1. If, $b=0.8$, multiplier $(m) = 1/(1 - 0.8) = 1/0.2 = 5$
2. If, $b=0.5$, multiplier $(m) = 1/(1 - 0.5) = 1/0.5 = 2$
3. If, $b=0.3$, multiplier $(m) = 1/(1 - 0.3) = 1/0.7 = 1.43$

The value of the 'autonomous spending multiplier' depends on assumptions in Keynesian model

Modified Keynesian Model - 2: Introduce income-based tax

- $C = 30 + 0.75 (Y - T)$
- $I_{\text{planned}} = 50$
- $G = 100$
- $NX = 30$
- $T = 0.2 Y$

What is the equilibrium value of output?

Modified Keynesian Model 2: Solution

- $C = 30 + 0.75 (Y - T)$

- $I_{\text{planned}} = 50$

- $G = 100$

- $NX = 30$

- $T = 0.2 Y$

$$PAE = Y_{\text{planned}} = 30 + 0.75 (Y - 0.2Y) + 50 + 100 + 30 = 210 + 0.6 Y$$

At equilibrium, $PAE = Y$ or, $Y = 210 + 0.6Y$ or, $Y = (1/0.4) * 210 = 2.5 * 210$

Thus, $Y^* = 525$

How does the multiplier modify in Keynesian model 2?

At equilibrium, $PAE = Y$

Thus, $Y = A + b(Y - tY)$ or $Y - bY - btY = A$ or $Y(1 - b(1 - t)) = A$ or

Or, $Y = 1/[(1 - b(1 - t))] * A$

With change in A by ΔA , we get,

$$\Delta Y = 1/[(1 - b(1 - t))] * (\Delta A) = m * (\Delta A)$$

Thus, in this model. $m = 1/[(1 - b(1 - t))]$

It is lower than $1/(1 - b)$ **why?**

Modified Keynesian Model - 3: Impact of income-based taxation and income induced imports:

- $C = 30 + 0.75 (Y - T)$
- $I_{\text{planned}} = 50$
- $G = 100$
- $NX = 80 - 0.1 Y$
- $T = 0.2Y$

Based on the above, what is the value of equilibrium output?

Modified Keynesian Model - 3: Solution

- $C = 30 + 0.75 (Y - T)$
- $I_{\text{planned}} = 50$
- $G = 100$
- $NX = 80 - 0.1 Y$, and $T = 0.2Y$

$$PAE = Y_{\text{planned}} = 30 + 0.75(Y - 0.2Y) + 50 + 100 + 80 - 0.1Y = 260 + 0.5 Y$$

At equilibrium, $PAE = Y$ or $Y = 260 + 0.5 Y$ or $Y = (1/0.5) * 260 = 2 * 260$

Thus, **$Y^* = 520$**

How does the multiplier modify in Keynesian model 3?

At equilibrium, $PAE = Y$

Thus, $Y = A + b(Y - tY) - dY$ or $Y - bY - btY + dY = A$ or $Y(1 - b(1 - t) + d) = A$

Or, $Y = 1/[(1 - b(1 - t) + d)] * A$

With change in A by ΔA , we get,

$$\Delta Y = 1/[(1 - b(1 - t) + d)] * (\Delta A) = m * (\Delta A)$$

Thus, in this model. $m = 1/[(1 - b(1 - t) + d)]$

It is lower than $1/(1 - b)$ and $1/[(1 - b(1 - t))]$

why?

Key learning points ...

- An increase in autonomous spending raises the equilibrium level of income.
- The increase in income is a multiple of the increase in autonomous spending.
- The larger the marginal propensity to consume, the larger the multiplier arising from the relation between consumption and income.

Discussion
